DO NOW — RECALL

Lesson 112 · Choosing Applications and Utility Software

Name: _____ Class: _____ Date: _____

1. What role does a device driver perform?

2. A linked-list node has data and what other essential field?

3. In a BST, where is a value less than the current node placed?

4. Why can an array provide direct indexed access?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It lets the OS communicate with/control a specific hardware device.

2. Model answer

A link/reference to the next node (for a singly linked list).

3. Model answer

In its left subtree (under the chosen no-duplicate convention).

4. Model answer

Its element locations can be calculated from base address and index.

By the end, you can…

• Identify users, scale, accessibility, platform and constraints.

• Select task-specific tools: DBMS, spreadsheet, CAD, DTP, etc.

• Backup, compression, disk tools, antivirus and encryption maintain/protect systems.

Vocabulary — know these terms

application software — Programs that meet a user task.

utility — Software for maintenance, protection or optimisation.

DBMS — Database management system.

backup — Recoverable copy to protect against data loss.

NEW LEARNING

Read, annotate and explain

Clarify needs

Identify users, scale, accessibility, platform and constraints.

Match applications

Select task-specific tools: DBMS, spreadsheet, CAD, DTP, etc.

Understand utilities

Backup, compression, disk tools, antivirus and encryption maintain/protect systems.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A clinic manages thousands of appointments accessed concurrently. Spreadsheet or database application?

Model answer

A DBMS is better: structured records, querying, concurrency, constraints and access rights. A spreadsheet is useful for small-scale analysis but is not ideal as a multi-user transactional store.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Justify an application for editing a school magazine with text and complex page layout. [6]

Question 2

Differentiate application software from utility software using two examples. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A small charity needs inventory records, backups and accessible user interfaces. Recommend software and justify limitations. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A backup utility and a word processor are both examples of utility software.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.